(Q1)

(a) Bandwidth $= 13 \times 10^3$ kHz $-$ 800 kHz

$= 12.2 \times 10^3$ kHz

$= 12.2 \times 10^6$ Hz

Data Rate $= \dfrac{108 \times 1024^3 \times 8}{8 \times 3600}$

$= \boxed{32.21 \times 10^6}$ bps.

$\therefore$ Nyquist Bit Rate $= 2 \times B \times \log_2 L$

$\Rightarrow 32.21 \times 10^6 = 2 \times 12.2 \times 10^6 \times \log_2 L$

$\Rightarrow L = 2.49 \approx 3$ Ans

(b)

Signal of noise $= 30$ mW

" " signal $= 20 \times 30 = 600$ mW

$\therefore$ SNR $= \dfrac{600}{30} = 20$

$\therefore$ Shannon Capacity $= B \log_2 (1 + SNR)$

$= 12.2 \times 10^6 \log_2 (1 + 20)$

$= 53.59 \times 10^6$ bps Ans

∴ Now,

$$10\log_{10}\left(\frac{\text{Tower 1}}{\text{Alice}}\right) + 10\log_{10}\left(\frac{\text{Satellite}}{\text{Tower 1}}\right) + 10\log_{10}\left(\frac{\text{Tower 2}}{\text{Satellite}}\right) +$$

$$10\log_{10}\left(\frac{\text{Bob}}{\text{Tower 2}}\right) = -0.9745 \text{ dB}$$

Now, Let,

power at Tower 2 = $960 - x$

∴ power at ~~Alice~~ Bob's phone = $2(960 - x)$

$$\therefore 10\log_{10}\left(\frac{2(960-x)}{P_{(Alice)}}\right) = -0.9745$$

$$\Rightarrow 10\log_{10}\left(\frac{2(960-x)}{801}\right) = -0.9745$$

$\Rightarrow x = 639.9 \text{ mW} \approx 640 \text{ mW}$

∴ Power at tower $= 960 - 640$ mW
$= 320$ mW

∴ " " Bob's phone = 640 mW

∴ Attenuation for Satellite to Tower 2 = $10\log_{10}\left(\frac{320}{960}\right)$

$= 10\log_{10} \frac{1}{3}$

$= -4.778$ dB

Q8

(i) Total bits = Data Rate × time $= 10 \times 10^6 \times 30$ bits

$\therefore$ Total characters $= \frac{10 \times 10^6 \times 30}{8} = 37.5 \times 10^6$ charact

(ii)

data rate = 10 Mbps $= 10 \times 10^6$ bps

Bandwidth = 1.5 MHz

$= 1.5 \times 10^6$ Hz

$\therefore$ Nyquist channel $= 2B \log_2 L$

$\Rightarrow 10 \times 10^6 = 2 \times 1.5 \times 10^6 \times \log_2 L$

$\Rightarrow L = 10.079 \approx 10$

Q9

Alice to tower 1,

power loss in 1 km = 2.14 mW

$\therefore$ " " " 150 km $= 2.14 \times 150$ mW

$= 321$ mW

$\therefore$ Remained ~~Attenuation~~, $A_1 = 801 - 321$

$= 480$ mW

Tower 1 to satellite,

$480 \times 2 = 960$ mW

let Sat to Tower 2's attenuation be A_2

$\therefore$ Tower 2 to Bob is $2 \times A_2$

Q10

(a) Bandwidth $= 119 - 104 = 15\ MHz = 15 \times 10^6\ Hz$

Signal - Noise Ratio $= \dfrac{22}{2} = 11$

∴ Theoretical Max, Nyquist Channel $= 2B \log_2 L$

∴ Shannon Capacity $= B \times \log_2(1+SNR)$

$= 15 \times 10^6 \times \log_2(1+11)$

$= 53774437.51$ bps

(b) Nyquist Channel $= 2B \cdot \log_2 L$

$\Rightarrow 53.78 \times 10^6 = 2 \times 15 \times 10^6 \times \log_2 L$

$\Rightarrow L = 3.46 \approx 4$

(c) $2 \times$ signal level $= 8$

→ Level increase leads to less reliability

→ $8 = 2^3 \Rightarrow 3$ bits per clock pulse

→ If noise effect, the possibility of বিকৃতি increases, creating problem.

$\therefore$ Shanon Capacity $= B \log_2 (1+SNR)$

$\Rightarrow 38.7 \times 10^6 = 68 \times 10^6 \log_2 (1+SNR)$

$\Rightarrow SNR = 0.4836$

$\therefore SNR_{dB} = 10 \log_{10} (SNR) = -3.155$ dB

<u>Q7</u>

Bandwidth, $p - q = 20$ kHz
$= 20 \times 10^3$ Hz

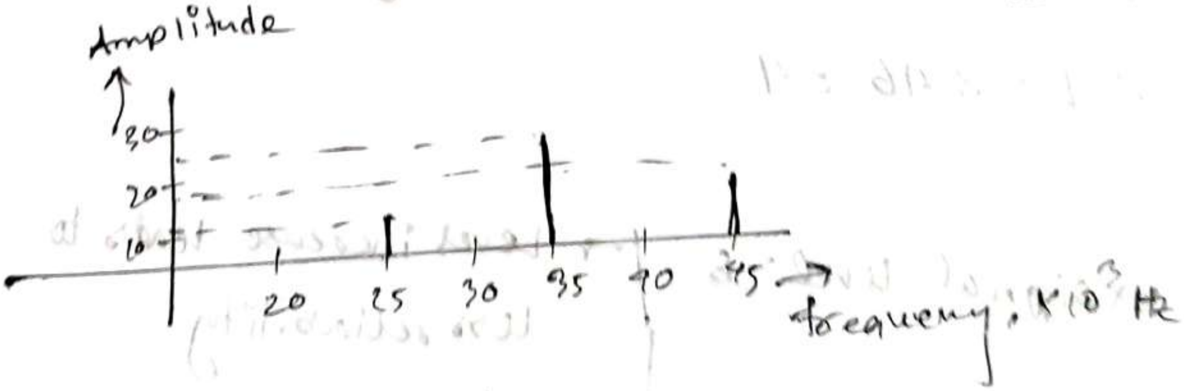
Minimum frequency $= 25$ kHz
$= 25 \times 10^3$ Hz

$\therefore p - q = 20 \times 10^3$
$q = 25 \times 10^3$

Max freq, $P = 45 \times 10^3$ Hz

Mid frequency $=$
$\frac{45+25}{2} \times 10^3$
$= 35 \times 10^3$ Hz

fq^n	Amp^d
45×10^3	18V
35×10^3	24V
25×10^3	12 V



(iii) Same datarate 4MHz bandwidth $= 4\times10^6$ Hz

$\therefore 2\times 4\times 10^6 \times \log_2 L = 7.9\times10^6$

$\Rightarrow L = 1.98 \approx 2$

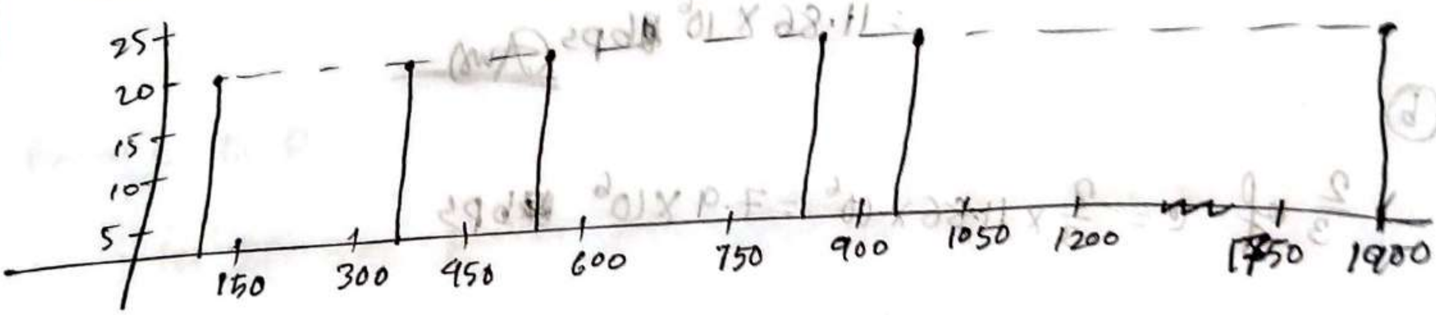
Q5

Six sine waves → 126, 348, 544, 896, 957, 1900 Hz.

Amplitude of all component = 20V

$\therefore$ Bandwidth = Max frequency − Min frequency

$= 1900 - 126 = 1774$ Hz



Q6

lower frequency, $p = 1863$ MHz

Higher " " $q = 1931$ MH

Bandwidth $= 1931 - 1863$

$= 68$ MHz

$= 68\times10^6$ Hz

$\therefore$ Upper Limit = Max bitrate = 38.7 Mbps

$= 38.7\times10^6$ bps

$\therefore$ change of power $= 10 \log_{10} \left(\frac{9033.89}{6000} \right)$

$= \underbrace{1.78 \text{ dB}}$

↳ amplified.

Q3

Bandwitth, $B = 2MHz = 2 \times 10^6$ Hz

Signal to Noise, Ratio, $SNR = 60$

$\therefore$ Shanon capacity $= B \times \log_2 (1+SNR)$

$= 2 \times 10^6 \times \log_2 (1+61)$

$= 11861474.69$ bps

$= 11.86 \times 10^6$ bps (Ans).

(b)

$\frac{2}{3}$ of $c = \frac{2}{3} \times 11.86 \times 10^6 = 7.9 \times 10^6$ bps

$\therefore C = 2B \log_2 L$

$\Rightarrow 7.9 \times 10^6 = 2 \times 2 \times 10^6 \times \log_2 L$

$\Rightarrow L = 3.931 \approx 4$ Ans

Q_2

From A to B,

in 1 km, loss = 3 kW

∴ " 100 " , " = 300 kW

$1\,MW = 10^6\,W$

$\Rightarrow 6\,MW = 10^6 \times 6\,W$

$\Rightarrow 1\,W = 10^{-3}\,kW$

$\Rightarrow 6\times10^6\,W = 6\times10^6\times10^{-3}\,kW$

signal at point A, $P_A = 6\,MW$

$= 6\times10^6\,W$

$= 6\times10^3\,kW$

∴ signal at point B, $P_B = 6\times10^3\,kW - 300\,kW$

$= 5700\,kW$

from BC, power amplification = $5700 \times 2 = 11400\,kW$ → P_c

From C to D,

for 200 km, power loss = $0.04\times200 = 8\,dB$

$$10\log_{10}(P_D/P_c) = -8\,dB$$

$$\Rightarrow \log_{10}(P_D/11400) = -0.8$$

$$\Rightarrow P_D = 11400\times10^{-0.8}$$

$$= 1806.77\,kW$$

∴ 5 time amplification = $1806.77\times5 = 9033.89\,kW$

Q1

from A to B,

distance = 0.5 km.

$\therefore$ in 1 km, power attenuation = -3.5 dB

$\Rightarrow$ " 0.5 " " " $= -3.5 \times 0.5 = -1.75$ dB

Again, from B to C,

distance = 0.6 km

$\therefore$ in 1 km, power attenuation = -4 dB

$\therefore$ " 0.6 " " " $= -4 \times 0.6$ dB $= -2.4$ dB

Again, from C to D,

distance = 0.5 km

$\therefore$ in 1 km, power attenuation = 8 dB

$\therefore$ " 0.5 km, " " $= 8 \times 0.5 = 4$ dB.

$\therefore$ Total attenuation $= -1.75 + 4 - 2.4 = \underbrace{-0.15}_{\text{power loss}}$ dB